

EXPERIMENT - 4

4. Implement histogram equalization on the given image and compare it with the original image using Open CV.

Aim

To implement Histogram Equalization on a given image and compare the equalized image with the original image using OpenCV.

Procedure

1. Import the OpenCV library.
2. Read the input image using cv2.imread().
3. Convert the image to grayscale.
4. Apply histogram equalization using cv2.equalizeHist().
5. Display the original grayscale image and the histogram-equalized image.
6. Display the histograms of both images for comparison.
7. Wait for a key press and close all windows.

Program

```
import cv2
import matplotlib.pyplot as plt

image = cv2.imread("image.jpg")

if image is None:
    print("Image not found!")
else:
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    equalized = cv2.equalizeHist(gray)

    cv2.imshow("Grayscale Image", gray)
    cv2.imshow("Histogram Equalized Image", equalized)

    plt.figure(figsize=(10, 4))

    plt.subplot(1, 2, 1)
```

```
plt.hist(gray.ravel(), 256, [0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(1, 2, 2)
plt.hist(equalized.ravel(), 256, [0, 256])
plt.title("Equalized Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input

Image.jpg



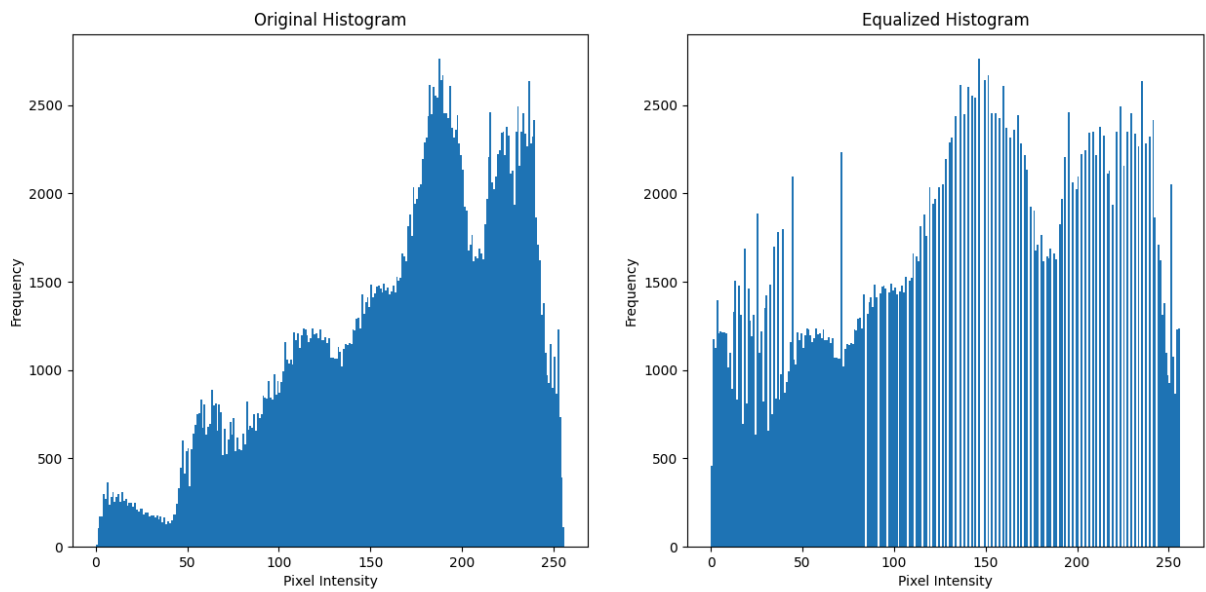
Sample Output

 Grayscale Image



 Histogram Equalized Image





Result

Thus, histogram equalization was successfully implemented using OpenCV, and the original image was compared with the histogram-equalized image.